Generalised Extreme Value (GEV) distribution

Parametrisation

The GEV distribution is defined through the cummulative distribution function

$$F(y;\eta,\tau,\xi) = \exp\left(-\left[1+\xi\sqrt{\tau s}(y-\eta)\right]^{-1/\xi}\right)$$

for

$$1+\xi\sqrt{\tau s}(y-\eta) > 0$$

and for a continuously response y where

η**:** is the linear predictor

τ**:** is the "precision"

s**:** is a fixed scaling, $s > 0$.

Link-function

The linear predictor is given in the parameterisation of the GEV distribution.

Hyperparameters

The GEV-models has two hyperparameters. The "precision" is represented as

$$\theta_1 = \log\tau$$

and the prior is defined on θ_1. The shape parameter ξ is represented as

$$\theta_2 = \xi$$

and the prior is defined on θ_2. [1]

Specification

- `family="gev"`
- Required arguments: y and s (keyword `scale`)
- The scaling ξ_s is given by the argument `gev.scale.xi` and is default set to 0.1.

The weights has default value 1.

[1] Internally, the parameter θ_2 is scaled with a fixed scaling ξ_s (default 0.1), to improve the numerics as the natural "scale" of ξ is small. For this reason the $\theta_2(=\xi)$ reported in `result$mode$theta` will appear as θ_2/ξ_s. For the same reason, if you define the mode using `control.mode = list(theta = ..., ...)` then the element representing θ_2 should be given as θ_2/ξ_s.

Hyperparameter spesification and default values

doc `The Generalized Extreme Value likelihood`

hyper

theta1

hyperid `76001`
name `log precision`
short.name `prec`
output.name `precision for GEV observations`
output.name.intern `log precision for GEV observations`
initial `4`
fixed `FALSE`
prior `loggamma`
param `1 5e-05`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

theta2

hyperid `76002`
name `tail parameter`
short.name `tail`
output.name `tail parameter for GEV observations`
output.name.intern `tail parameter for GEV observations`
initial `0`
fixed `FALSE`
prior `gaussian`
param `0 25`
to.theta `function(x) x`
from.theta `function(x) x`

survival `FALSE`

discrete `FALSE`

link `default identity`

status `disabled: Use likelihood model 'bgev' instead; see inla.doc('bgev')`

pdf `gev`

Example

In the following example, we estimate the parameters of the GEV distribution on some simulated data.

```
rgev = function(n=1, xi = 0, mu = 0.0, sd = 1.0) {
    u = runif(n)
    if (xi == 0) {
        x = -log(-log(u))
    } else {
```

```
        x = ((-log(u))^(-xi) - 1.0)/xi
    }
    return (x*sd + mu)
}

n = 300
z = rnorm(n)
sd.y = 0.5
xi = 0.2
y = 1+z + rgev(n, xi=xi, sd = sd.y)

r = inla(y ~ 1 + z, data = data.frame(y, z), family = "gev",
        control.family = list(gev.scale.xi = 0.01))
summary(r)
```

Notes

None.